

HYDERABAD, Pakistan

WMO: 417640

Lat: 25.383N

Lon: 68.417E

Elev: 41

StdP: 100.83

Time Zone: 5.00 (E05)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	11.7	-5.0	2.5	22.5	-2.2	3.1	21.9	10.3	19.2	9.3	19.4	1.6	0	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.5	42.5	23.9	41.0	24.0	39.7	24.3	28.6	35.5	28.1	34.9	27.7	34.3	6.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.1	31.3	26.7	22.4	31.1	26.2	21.8	30.9	93.3	35.7	91.2	34.6	89.5	34.4	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	14.7	13.0	DB	8.1	45.5	2.2	0.8	6.6	46.1	5.3	46.5	4.1	47.0	2.5	47.5	(4)
			WB	4.8	30.2	2.4	1.2	3.1	31.0	1.7	31.7	0.3	32.3	-1.5	33.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	18.1	21.3	26.9	31.5	33.8	33.8	32.4	31.1	31.0	29.9	25.2	19.6
	DBStd	5.75	2.21	3.18	3.16	1.91	1.43	1.46	1.40	1.23	1.41	1.74	2.32	2.52
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	57	32	9	1	0	0	0	0	0	0	0	0	15
	CDD10.0	6538	251	315	522	646	739	713	696	653	630	617	457	299
	CDD18.3	3554	25	91	264	396	481	463	437	395	380	359	207	56
	CDH23.3	46835	160	742	2986	5480	7177	7006	6273	5249	5044	4511	1854	353
	CDH26.7	27241	15	245	1606	3406	4717	4612	3799	2788	2722	2487	775	68
Wind	WSAvg	5.1	2.7	3.0	3.5	4.9	7.7	8.6	8.7	7.6	5.9	3.2	2.6	2.9
Precipitation	PrecAvg	166	2	4	4	6	3	7	50	55	19	2	2	1
	PrecMax	487	12	16	47	47	47	75	213	285	286	25	34	29
	PrecMin	5	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	136	3	5	10	12	8	17	58	65	50	5	7	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.2	33.7	40.2	43.1	44.6	44.3	40.9	38.7	39.3	39.8	35.7	30.5
		MCWB	17.0	18.6	20.4	21.5	24.2	24.1	26.9	26.7	24.4	21.2	19.7	18.6
	2%	DB	26.6	31.8	37.8	41.2	42.9	41.5	38.5	36.8	37.8	38.1	33.9	28.7
		MCWB	16.5	18.2	20.2	21.9	24.0	26.1	27.2	26.7	25.2	21.8	19.7	17.6
	5%	DB	25.1	29.8	35.9	39.9	41.4	39.7	37.3	35.6	36.5	36.8	32.5	27.0
		MCWB	15.5	17.7	19.7	21.7	24.4	26.7	27.2	26.7	25.5	21.8	19.3	16.9
	10%	DB	23.6	27.9	34.2	38.5	40.0	38.5	36.3	34.6	35.2	35.5	31.0	25.3
		MCWB	14.8	16.9	19.4	21.9	24.7	26.8	27.2	26.6	25.5	21.8	18.6	15.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	21.6	24.0	27.4	29.3	29.2	29.1	28.6	28.4	26.7	23.7	20.3
		MCDB	25.8	28.5	34.1	37.6	39.0	36.8	35.7	33.9	34.3	33.9	29.9	26.7
	2%	WB	18.0	20.5	22.7	25.9	28.0	28.4	28.4	28.1	27.7	26.0	22.5	19.0
		MCDB	23.7	27.4	30.7	35.0	36.7	36.0	34.9	33.5	33.3	32.6	28.4	25.9
	5%	WB	16.7	19.6	21.9	25.1	27.3	28.0	28.0	27.6	27.1	25.4	21.6	18.1
		MCDB	23.1	26.5	29.9	32.9	35.4	35.5	34.4	33.1	32.9	31.5	28.1	24.7
	10%	WB	15.7	18.6	21.1	24.4	26.7	27.6	27.6	27.2	26.6	24.8	20.6	17.1
		MCDB	22.0	25.3	29.2	31.8	34.4	34.8	33.8	32.6	32.3	30.4	27.7	23.7
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.8	13.1	14.3	13.5	10.5	8.4	7.8	9.2	11.7	11.2	10.5
		MCDBR	12.7	14.1	15.5	16.2	15.5	12.6	9.8	9.3	10.9	13.3	13.1	12.5
		MCWBR	5.3	4.7	4.2	4.2	3.6	2.5	2.0	2.3	2.5	3.4	3.8	4.6
	5% WB	MCDBR	12.0	12.4	14.1	14.8	13.7	10.6	8.7	8.0	8.9	11.1	11.1	11.4
	MCWBR	5.8	4.3	4.4	4.0	3.5	2.4	2.3	2.5	2.4	3.0	3.9	4.8	
Clear-Sky Solar Irradiance	taub	0.459	0.474	0.534	0.594	0.645	0.719	0.780	0.715	0.614	0.559	0.551	0.513	
	taud	1.953	1.951	1.817	1.727	1.669	1.570	1.510	1.611	1.765	1.788	1.760	1.824	
	Ebn at Noon	773	798	772	737	699	645	606	644	699	710	675	695	
	Edh at Noon	164	176	212	237	251	275	292	263	220	205	198	180	
All-Sky Solar Radiation	RadAvg	4.12	4.99	5.99	6.63	6.93	6.59	5.70	5.62	5.76	5.33	4.31	3.89	
	RadStd	0.17	0.19	0.19	0.18	0.12	0.27	0.36	0.26	0.29	0.21	0.18	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	N/A	-0.48	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page